

O-band Membrane Surface-Emitting Laser on a Si Substrate Demonstrating 100-Gbps PAM-4 Operation

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Abstract: We demonstrate an O-band surface-emitting laser on a Si substrate, based on an in-plane membrane distributed-reflector laser with a second-order surface grating, exhibiting a high modulation efficiency and 100-Gbps PAM-4 operation over 2-km SSMF. © 2025 The Authors

1. Introduction

Surface-emitting lasers (SELs) in the O/C-bands hold great potential for short- to long-reach interconnects that meet the ever-increasing datacenter traffic demands, owing to their two-dimensional (2D) ultra-dense integrability and the low-loss, low-dispersion transmission advantages of these bands. While considerable progress has been made in the O/C-band VCSELs [1-4], they still face challenges inherent to InP-based materials [1,2]. Specifically, their modulation bandwidths typically remain around 10 GHz [1,2], with the highest reported values of 17 GHz [3] and 21 GHz [4] for the O- and C-bands, in contrast to the GaAs-based shorter-band VCSELs reaching up to 50 GHz [5].

In this context, we recently proposed “membrane surface-emitting lasers” [6], comprising a thin InP layer with first- and second-order surface gratings acting as a distributed Bragg reflector (DBR) and a grating coupler (GC) for in-plane cavity formation and grating-coupled surface emission, respectively. The membrane structure surrounded by low refractive index materials exhibits strong optical confinement into the buried active region, which is a key feature for high-efficiency, high-speed direct modulation; our membrane edge-emitting lasers (EELs) have achieved modulation bandwidths of ~25 GHz on Si substrates [7,8] and up to 108 GHz on SiC substrates with a photon-photon-resonance (PPR) effect [9]. The membrane structure with surface gratings provides a high coupling constant and a high out-of-plane radiation constant for the first- and second-order gratings, respectively, enabling short-cavity lasers with low power consumption [10] and high-efficiency grating-coupled surface emission [6].

In [6], we fabricated membrane SELs on a six-pair TiO₂/SiO₂ bottom reflector formed on a Si substrate to ensure high bottom reflectance. However, the >2-μm-thick reflector impeded efficient heat dissipation to the Si substrate, which limited the maximum injection current and thus hindered high-speed direct modulation. In this work, we improve the heat dissipation by reducing the distance between the laser and the Si substrate, through a newly designed ~560-nm-thick amorphous-Si (a-Si)/SiO₂ intermediate reflective structure that maintains a high upward radiation efficiency. The device exhibits a 9.1-GHz/ma^{1/2} modulation current efficiency and a 25-GHz modulation bandwidth, and demonstrates 100-Gbps PAM-4 operation over 2-km standard single-mode fiber (SSMF).

2. Device Design and Fabrication

Figure 1(a) schematically illustrates the membrane SEL designed in this work, whose active section comprises a GC

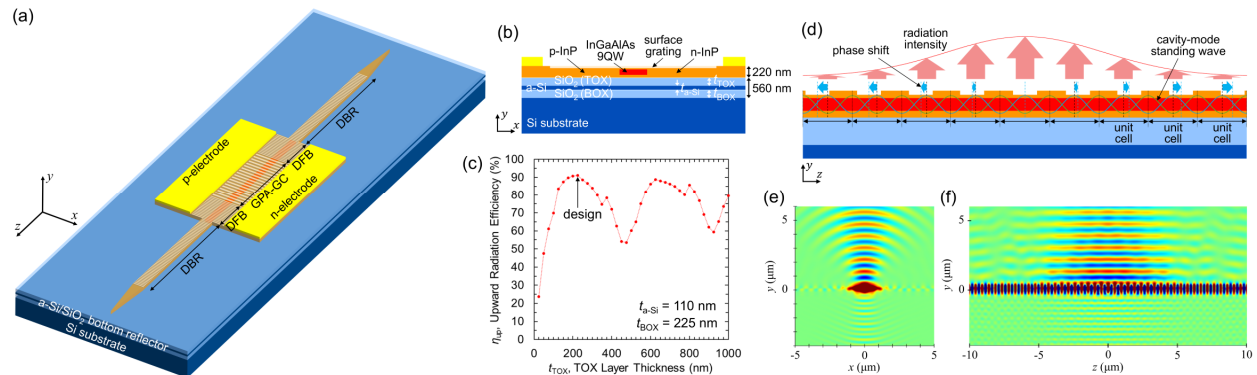


Fig. 1. (a) Schematic overview of the membrane SEL on a Si substrate. (b) x-y cross-section showing the layer structure. (c) η_{up} of the membrane GC versus t_{TOX} . (d) y-z cross-section showing the grating, cavity-mode, and surface-emission profiles of the GPA-GC section. (e), (f) FDTD-calculated electric-field radiation profiles of the SEL cavity with 9.2-μm target MFD: (e) $\text{Re}[E_x](x,y)$ and (f) $\text{Re}[E_x](y,z)$.

section sandwiched between distributed-feedback (DFB) sections. The active section is further sandwiched between passive DBR sections, forming a distributed-reflector (DR) laser structure. As shown in Fig. 1(b), the active sections consist of a 220-nm-thick InP membrane containing a buried active region made of InGaAlAs-based nine quantum wells with a lateral-current-injection (LCI) structure, while the passive sections consist of InP channel waveguides butt-coupled to the active section. The intermediate stack between the InP membrane and Si substrate consists of a top SiO₂ (TOX) layer, an a-Si layer, and a buried SiO₂ (BOX) layer with thicknesses of t_{TOX} , $t_{\text{a-Si}}$, and t_{BOX} , respectively. The a-Si and BOX thicknesses were set by the quarter-wave condition $t = \lambda/(4n)$ using the measured refractive index of a-Si ($n_{\text{a-Si}} = 2.97$), yielding $t_{\text{a-Si}} = 110$ nm and $t_{\text{BOX}} = 225$ nm. The upward radiation efficiency η_{up} of the membrane GC structure was calculated as a function of t_{TOX} using the finite-difference time-domain (FDTD) method, as shown in Fig. 1(c). A maximum η_{up} of 91% was obtained at $t_{\text{TOX}} = 225$ nm, even though the bottom reflectance R_{bot} of the a-Si/BOX/Si-substrate stack was only ~67%. The oscillatory dependence on t_{TOX} indicates that constructive interference is harnessed to obtain a high η_{up} even with a moderate R_{bot} . In combination with employing a-Si, which has a higher refractive index than TiO₂, this design allows a reduction in the bottom-reflector thickness from [6] while maintaining a high η_{up} .

The detailed design of the longitudinal cavity structure is shown in Fig. 1(d). Between the DFB sections lies our proposed “Gaussian-phase-apodized grating coupler” (GPA-GC) [6], in which the grating phase profile is apodized to produce a Gaussian radiation profile with a desired mode-field diameter (MFD), with an appropriate cavity-mode standing wave in phase with the GC unit cells. When the MFD is set well below the GPA-GC length, this design ensures single-mode lasing of the Gaussian-radiating mode, because other modes radiate throughout the GPA-GC section and thus incur much higher losses. The target MFD is set to 9.2 μm to facilitate SSMF coupling. Figures 1(e) and (f) present FDTD-calculated radiation profiles for the SEL cavity with a 40- μm -long GPA-GC section, confirming the intended Gaussian-like upward beam with only minor downward leakage into the Si substrate.

For heterogeneously fabricating the membrane SELs, we employed the micro-transfer-printing method [8]; the membrane SEL structures were fabricated on an InP substrate, with an underlying sacrificial layer and an ~400-nm-thick SiON overcladding layer, whereas the BOX, a-Si, and TOX layers were deposited on a Si substrate to realize the designed bottom-reflector structure. Subsequently, through resist-tether formation and sacrificial-layer removal, the individual SEL coupons were transfer-printed onto the TOX surface of the Si substrate.

3. Device Characterization and Eye Measurement

We measured a fabricated membrane SEL with a 40- μm -long GPA-GC section, 15- μm -long DFB sections (total active length $L_a = 70$ μm), and 40- μm -long DBR sections terminated with InP inverse tapers. All measurements were performed at a stage temperature of 25°C. Figure 2(a) shows the L-I-V characteristics measured with the spatial photodiode (PD) setup, where a spatial PD with 9.5-mm- ϕ input aperture was placed ~5 mm above the membrane SEL to detect upward output power. A threshold current of 1.6 mA, a near-threshold slope efficiency of 0.29 W/A (inset of Fig. 2(a)), and a maximum output power of 1.88 mW at 13 mA were obtained. This slope efficiency is notably higher than those of buried-heterostructure grating-coupled SELs (e.g., 0.075 W/A [11]). Further increase in the slope efficiency is expected by optimizing the grating etch depth to obtain a higher radiation constant in the GC section. Figure 2(b) shows lasing spectra measured with the SSMF setup, where the cleaved facet of an SSMF (SMF-28e+, Corning) was vertically coupled to the GPA-GC section. Single-mode lasing was obtained over the entire current range with a high side-mode suppression ratio (SMSR) of 54 dB at 13 mA, indicating that the DR cavity stably supported an appropriate cavity mode. The SSMF coupling efficiency was measured to be -9.6 dB, which we expect to improve by integrating a cylindrical microlens onto the GPA-GC to mitigate the large x-axis

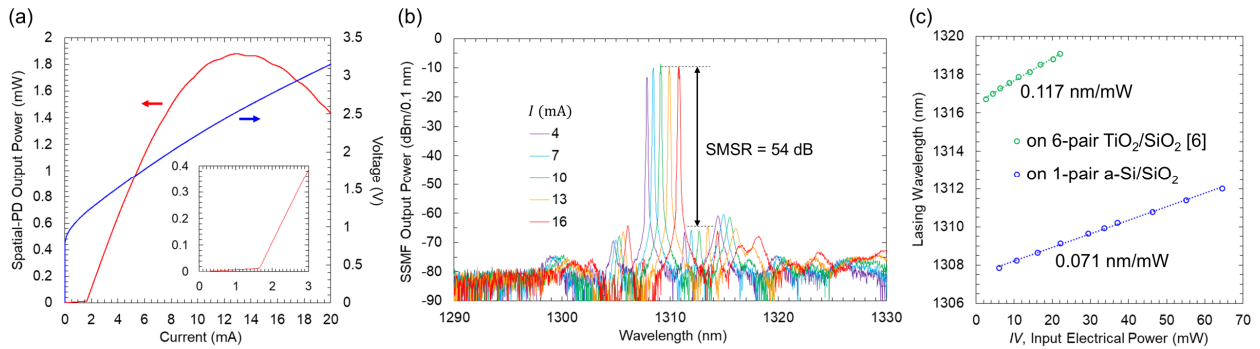


Fig. 2. (a) L-I-V characteristics of the fabricated membrane SEL at 25°C measured with the spatial PD setup. The inset shows a close-up view around the lasing threshold. (b) Lasing spectra at 25°C measured with the SSMF setup. (c) Lasing wavelength as a function of input electrical power for the present device and for the device on the six-pair TiO₂/SiO₂ bottom reflector fabricated in [6].

MFD mismatch (Fig. 1(e)). Figure 2(c) shows lasing wavelength versus input electrical power for the present device and for the device on a six-pair $\text{TiO}_2/\text{SiO}_2$ bottom reflector [6]. With the same active length L_a of 70 μm , the former exhibits a self-heating wavelength-shift coefficient of 0.071 nm/mW, which is $\sim 40\%$ lower than that of the latter. This confirms improved heat dissipation thanks to the use of the a-Si/SiO₂ bottom reflector.

Finally, we performed dynamic characterization and eye-diagram measurement at 25°C, by using the SSMF setup, with a praseodymium-doped fiber amplifier (PDFA) to compensate for the SSMF coupling loss. Figure 3(a) shows the electro-optic (E/O) small-signal responses, exhibiting a maximum 3-dB modulation bandwidth $f_{3\text{dB}}$ of 25.0 GHz at 15 mA. To the best of our knowledge, this modulation bandwidth represents a record-high value for O/C-band SELs. Figure 3(b) plots $f_{3\text{dB}}$ from the E/O responses and the relaxation oscillation frequency f_r from the relative intensity noise (RIN) measurement, versus square root of the bias current above threshold, yielding a modulation current efficiency factor (MCEF) of 9.1 GHz/mA^{1/2} and a D-factor of 7.1 GHz/mA^{1/2}. Figures 2(c) and (d) present non-return-to-zero (NRZ) eye diagrams at 40 and 50 Gbps for a 2³¹-1 pseudo-random bit sequence (PRBS) generated by a pulse-pattern generator (PPG) under a back-to-back (BTB) setup without any equalization. The bias current and the signal's peak-to-peak voltage were 13 mA and 0.6 V, respectively. Clear eye openings with extinction ratios (ERs) over 2 dB confirm 50-Gbaud-class modulation capability. Figures 2(e)-(h) show PAM-4 eyes at 80 and 100 Gbps for a 2¹³-1 PRBS generated by an arbitrary-waveform generator (AWG), evaluated in BTB and 2-km SSMF transmission setups, using a 5-tap feed-forward equalizer (FFE) in the receiver sampling oscilloscope. The bias current and the signal's peak-to-peak voltage were 13 mA and 1.2 V, respectively. All cases exhibit eye openings with outer ERs exceeding 4 dB, demonstrating up to 100-Gbps PAM-4 operation over 2-km SSMF. At this bias current, the device consumes 32.9 mW, yielding energy costs of 411 fJ/bit and 329 fJ/bit for the 80- and 100-Gbps PAM-4 operations, respectively.

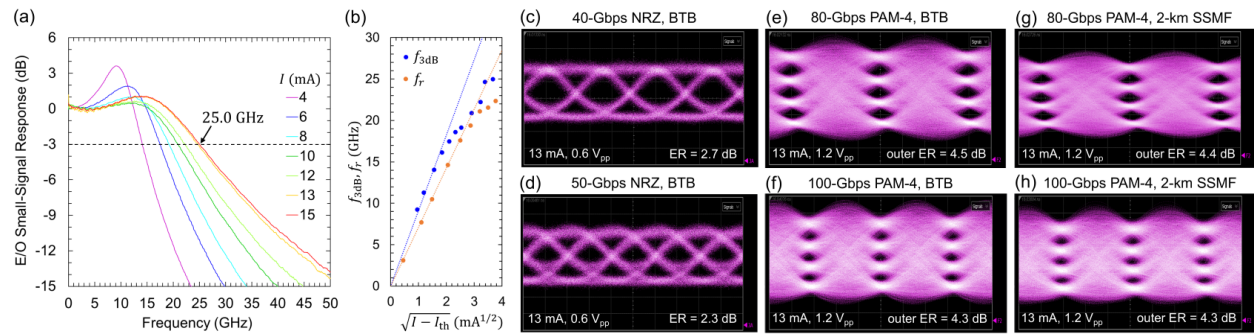


Fig. 3. Dynamic characteristics and eye diagrams of the membrane SEL at 25°C; (a) E/O small-signal responses. (b) $f_{3\text{dB}}$ and f_r versus square root of the bias current above threshold. (c),(d) 40- and 50-Gbps NRZ eye diagrams without equalization in BTB. (e),(f) 80- and 100-Gbps PAM-4 eye diagrams with a 5-tap FFE in BTB. (g),(h) 80- and 100-Gbps PAM-4 eye diagrams with a 5-tap FFE after 2-km SSMF transmission.

4. Conclusion

In summary, we have demonstrated an O-band membrane distributed-reflector SEL on a Si substrate incorporating a thin a-Si/SiO₂ intermediate layer that effectively serves as a bottom reflector and a heat sink. The device achieves an MCEF of 9.1 GHz/mA^{1/2}, reaches an $f_{3\text{dB}}$ of 25 GHz, and demonstrates 100-Gbps PAM-4 transmission over 2-km SSMF. These results highlight the high-modulation-efficiency advantage of membrane SELs and their promise as high-speed, 2D-arrayable transmitters for ultra-high-capacity short- to long-reach optical interconnects.

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